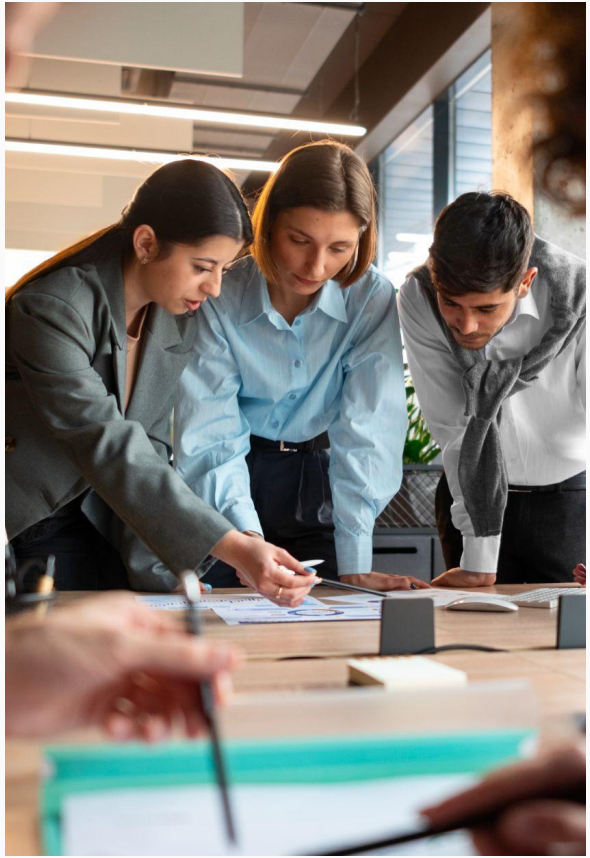
Data-Driven Strategy: Aligning ER Games with Industry Trends, Market Selection, and Consumer Preferences

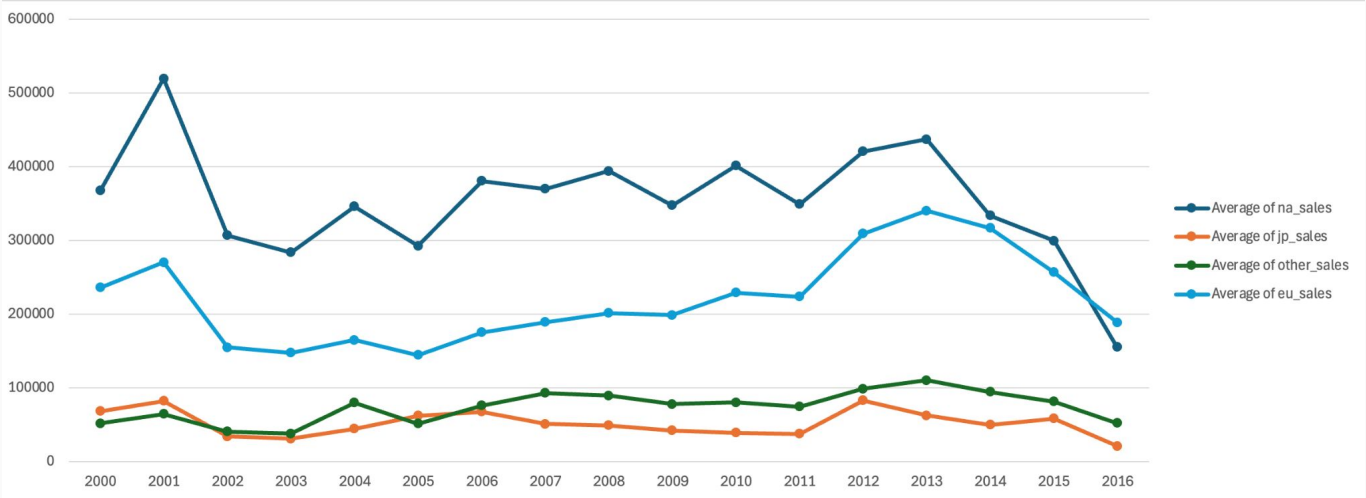
TEAM 11



01

Aligning ER Games with Industry Trends (KPI/Dashboarding)

Which market besides the US should ER Games target when releasing the game?



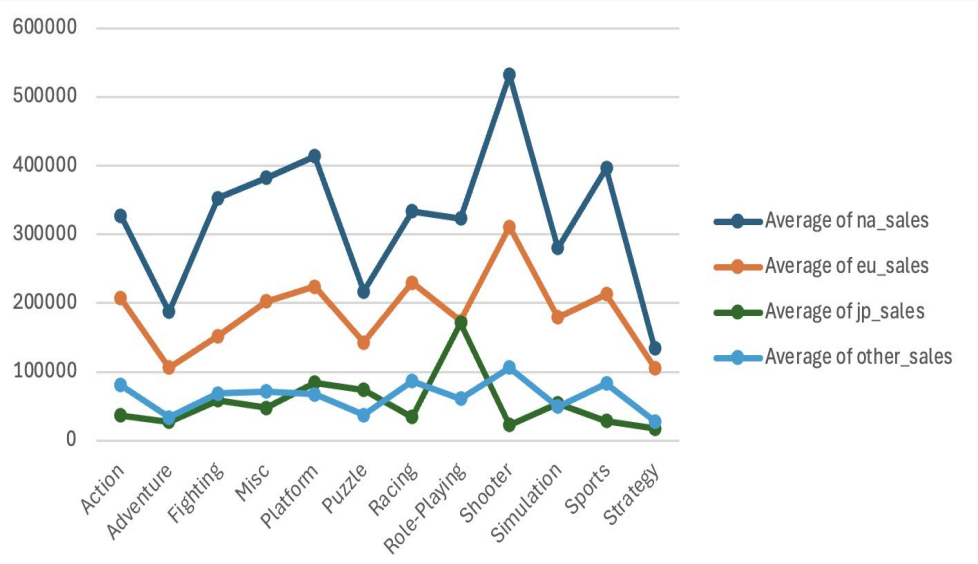
Insight

Identify the most successful game genres from 2000-2016

Findings

- North America (NA) is strongest market for video games across all genres, dominates in total sales (~354K).
- Europe (EU) is the second-best market (~206K), significantly ahead of Japan (~49K) and Other regions (~74K).

Market Trends - sales performance by genre across different markets



Insight

Identify the most successful game genres across different regions and whether ER Games newest games (hybrid action-strategy) aligns with market demand.

Findings

- Japan is the weakest market but find Role-Playing Games dominate in Japan (~171K sales), far surpassing Action (~36K) and Strategy (~17K).
- Action and Strategy games perform best in North America (~327K & ~134K) compared to Europe (~207K & ~105K), Japan (~36K & ~17K), and other regions (~80K & ~27K).

Conclusion

Chaderson and ER consultants were incorrect as the data suggests that Strategy games perform poorly in Japan and Europe, meaning these markets are not ideal for an action-strategy hybrid launch. **Asia & Europe should NOT be priority launch markets.**

Action vs. Strategy

Findings

Action games sell significantly better than Strategy games across North America (NA), Europe (EU), and other global markets.

NA has the highest sales for both Action and Strategy games, but Action games outperform Strategy by a large margin.

Recommendation

Launch the game in North America first

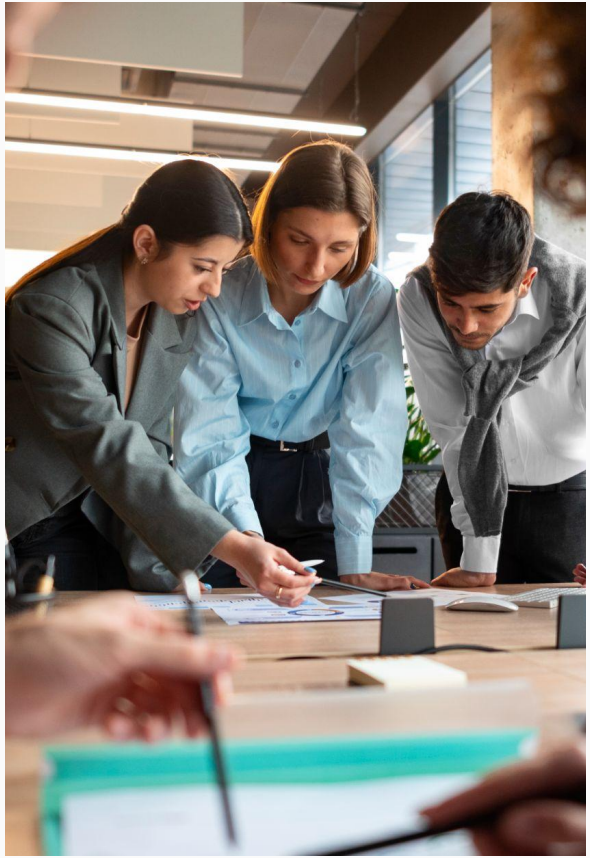
Both Action and Strategy games sell significantly better in NA compared to other regions which make it the best primary market.
Limitation: Without strong competition from major franchises, it might be difficult for ER Games to capture market share

Expand into Europe for max. impact

Europe have the best sales potential outside the U.S.
Risk mitigation strategy: Delay expansion into Japan and other regions until the game proves successful in its primary markets.

Position the game as an Action-heavy experience

Since ER Games is developing an Action-Strategy hybrid, the Action component should be the primary selling point to attract more players as strategy games have a smaller audience.



02

Market Selection (Clustering)

Values	Clusters 1	2	3	4	5
Average of na_sales	41510000	10593000	2054533	315095	155465
Average of eu_sales	29297000	7506727	1514431	165636	73514
Average of jp_sales	3795000	3345636	491382	33502	6937
Average of other_sales	8506000	3645091	534721	56859	27812
Average of critic_score	7.50	8.24	8.23	7.50	5.28
Average of user_score	7.90	8.09	7.73	7.67	5.38

Insight

We used hierarchical clustering to find the best way to group the data. There was a significant difference between 4 and 5 clusters, but only a small difference between 5 and 6 clusters. This confirmed that using 5 groups was the best choice.

This clustering helps us spot trends in the market, see which games perform well, and understand sales and review differences across regions.

Findings

1. **High Reviews Don't Guarantee High Sales** → Cluster 2 has the highest critic (8.24) & user (8.09) scores, meaning these games were well-received. However, their sales are lower, suggesting that strong reviews alone don't drive commercial success.
2. **Popular Games Don't Always Have the Best Reviews** → Cluster 1 has the highest sales but only moderate critic (7.5) & user (7.9) scores. This means that marketing and audience reach matter more for sales than reviews
3. **Poor Reviews Lead to Poor Sales** → Cluster 5 has the lowest critic (~5.28) & user (~5.38) scores, showing that these games failed both critically and commercially. ER Games should avoid designing games similar to Cluster 5.

Recommendation

If ER Games Focus on Commercial Success Over High Reviews

(Drawing on other business perspectives to assess implementation issues for the business)

ER Games should model its game after **Cluster 1**, where high sales matter more than perfect reviews. Games like Wii Sports, released in 2006, became bestsellers despite having mediocre reviews. This could be attributed to its marketing and brand. **Wii Sports** came bundled with all consoles besides japan. A strong word-of-mouth campaign in North America and Europe could boost sales.

name	platform	year_of_release	genre	publisher	na_sales	eu_sales	jp_sales	other_sales	critic_score	critic_count	user_score	user_count	developer	rating	HCluster
Weii Sports	Weii	2006	Sports	Ninti	41510000	29297000	3795000	8506000	7.5	51	7.9	322	Ninti	E	1

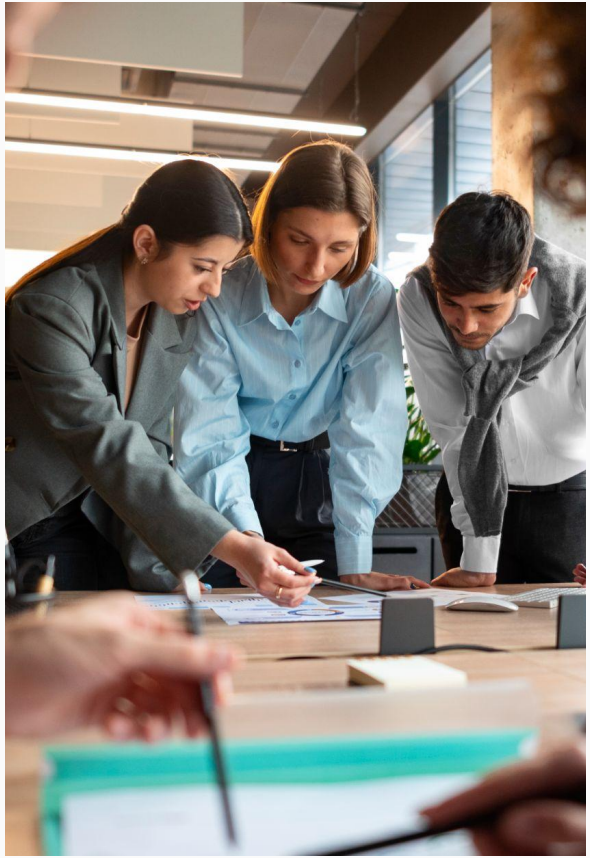
Recommendation

If ER Games wants to be known for quality, target critical acclaim

(Drawing on other business perspectives to assess implementation issues for the business)

Cluster 2 is the ideal model (higher reviews, but lower sales). This means investing in better game design and storytelling, even if it means slightly lower commercial success.

name	platform	year_of_release	genre	publisher	na_sales	eu_sales	jp_sales	other_sales	critic_score	critic_count	user_score	user_count	developer	rating	HCluster
Gran Turismo 4	PS2	2004	Racing	Sony Comput	3043000	10000	1112000	7439000	8.8	74	8.5	272	Polyphony Dig	E	2
Marrison Kart Weii	Weii	2008	Racing	Ninti	15888000	12517000	3719000	3247000	8.3	73	8.4	709	Ninti	E	2
Weii Sports Resort	Weii	2009	Sports	Ninti	15771000	11010000	3232000	2945000	8.1	73	8	192	Ninti	E	2
New Super Marrison Bros.	DS	2006	Platform	Ninti	11491000	9165000	6475000	2827000	8.7	65	8.6	431	Ninti	E	2
Weii Play	Weii	2006	Misc	Ninti	13847000	9192000	2929000	2829000	5.7	41	6.7	129	Ninti	E	2
New Super Marrison Bros. Weii	Weii	2009	Platform	Ninti	14713000	7066000	4647000	2215000	8.6	80	8.3	594	Ninti	E	2
Marrison Kart DS	DS	2005	Racing	Ninti	9752000	7609000	4117000	1901000	9.3	64	8.5	464	Ninti	E	2
Weii Fit	Weii	2007	Sports	Ninti	8834000	7990000	3530000	2163000	8	63	7.6	146	Ninti	E	2
Brain Age: Train Your Brain in Minutes a Day	DS	2005	Misc	Ninti	4734000	9034000	4128000	2076000	7.7	58	7.9	50	Ninti	E	2
Weii Fit Plus	Weii	2009	Sports	Ninti	9176000	8574000	2501000	1764000	8	33	7.5	52	Ninti	E	2
Grand Steal Auto: San Andreas	PS2	2004	Action	Take-Two Inte	9274000	407000	412000	10690000	9.4	80	9	1588	Rockstar Nort	M	2



02

Consumer Preferences (A/B Testing)

Impact of Game Genre on Popularity

	Stratergy vs. Sports	Strategy vs. Shooter
Diff	262368.0734	397987.6315
Z-score	6.401518871	9.63633232
p-value	0.00000000769193974560008%	0.00000000000000000000281%
Decision (Sig level 1%)	Sports	Shooter

Insight

We tested whether switching Strategy games to the two most popular game genres in North America would make a significant difference in performance.

Findings

- The decision at a 1% significance level shows Shooter and Sports games are significantly more popular than Strategy games (this means the difference in popularity between Strategy and the other genres is not due to random chance and is meaningful enough to switch)
- Strategy vs. Sports: Z-score: 6.40 → Strong statistical significance
- Strategy vs. Shooter: Z-score: 9.63 → Even stronger statistical significance
- **WHY?**

The Hidden Influence: Confounding Variables in Genre Performance

E	Stratergy vs. Sports	Strategy vs. Shooter	Decision (Sig level 1%)
Diff	201216.8732	-141357.1429	
Z-score	3.308894871	-3.387083773	
p-value	0.0468325001166073000000%	99.9646800928812000000000%	Sports
E10+	**Stratergy vs. Sports**	**Strategy vs. Shooter**	
Diff	205309.596	69317.47967	
Z-score	3.420760418	1.29692487	
p-value	0.0312231587556147000000%	9.73285192293723000000000%	Sports
M	**Stratergy vs. Sports**	**Strategy vs. Shooter**	
Diff	106221.7391	566748.0766	
Z-score	1.413893756	8.373436051	
p-value	7.86965497449069000000000%	0.00000000000000027981030%	Shooter
T	**Stratergy vs. Sports**	**Strategy vs. Shooter**	
Diff	213235.14	265576.8116	
Z-score	4.529223532	6.306006539	
p-value	0.000296004210282973000 0%	0.000000014316347145241 2%	Sports or shooter

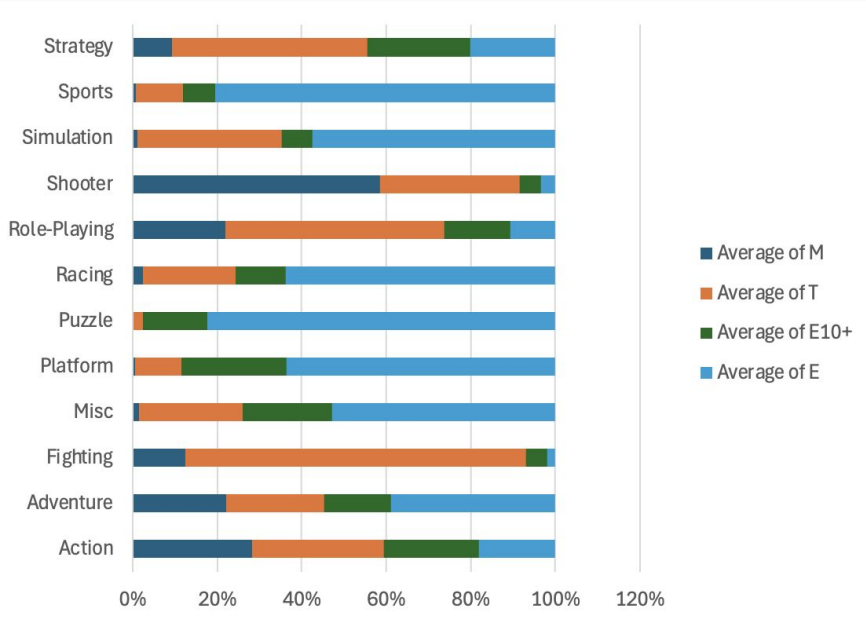
Finding

- Shooter games perform better than Strategy, but this varies by rating.
- In E & E10+ ratings, Sports is significantly more popular than Strategy (younger players prefer Sports over Strategy)
- In M-rated games, Shooters perform significantly better than Strategy (older players prefer intense experiences)
- T-rated games show no clear dominance between Sports & Shooter genres over Strategy.
- This suggests that T-rated Strategy games appeal to a broader audience compared to other ratings.

Conclusion

Confounding analysis shows that popularity/preference of ER games depends on the ESRB rating

Understanding Genre Preferences Through ESRB Ratings



Insight

We analyzed game preferences based on demographics by examining ESRB ratings across different genres.

Findings

- Family-friendly games (Puzzle, Racing, Platform) tend to be rated E.
- More intense games (Shooter, RPG, Fighting) are often rated T or M.
- Strategy games have a mix of ratings, suggesting a wide range of complexity and themes.
- Action games have a diverse mix of ratings, indicating the genre includes both kid-friendly and mature titles.

Conclusion

Strategy games perform well might because of accessibility. Sports and Shooter games are popular BUT they cater to more specific audiences, which can limit their overall advantage.

Future Recommendations

Develop an Action-Sports Hybrid or Action-Shooter

Action games have the highest sales (most popular genre)
Shooter games perform well across multiple ESRB ratings, attracting both younger and older players. Prioritize Sports mechanics over Strategy for younger audiences and Shooter mechanics for mature audiences. A hybrid of approach could **maximize both market size and engagement** by attracting more players

Limitation: Mixing game styles can be tricky, and if not done well, the game might not strongly appeal to either audience.

Risk Mitigation Strategy: Gather player feedback through before launch.

Thanks!